

L1.7 In-class Assignment – Matrices and Arrays

CPSC 292 – 9/9/2026

Name: _____ Section: _____

Collaboration team: _____

Question 1

Predict R's output for the following code:

```
matrix(1:40, nrow = 4, ncol = 10, byrow = FALSE)
```

Question 2

Write code that will replicate the array in Fig 1:

```
, , 1
      [,1] [,2] [,3] [,4] [,5]
[1,]    0   18   36   54   72
[2,]    3   21   39   57   75
[3,]    6   24   42   60   78
[4,]    9   27   45   63   81
[5,]   12   30   48   66   84
[6,]   15   33   51   69   87

, , 2
      [,1] [,2] [,3] [,4] [,5]
[1,]   90  108  126  144  162
[2,]   93  111  129  147  165
[3,]   96  114  132  150  168
[4,]   99  117  135  153  171
[5,]  102  120  138  156  174
[6,]  105  123  141  159  177

, , 3
      [,1] [,2] [,3] [,4] [,5]
[1,]  180  198  216  234  252
[2,]  183  201  219  237  255
[3,]  186  204  222  240  258
[4,]  189  207  225  243  261
[5,]  192  210  228  246  264
[6,]  195  213  231  249  267
```

Figure 1: Array in Question

Question 3

The array in Fig. 1 is stored as `my_array`. Overwrite the third column of the second matrix in this array with the same numbers in decreasing order.

Question 4

Reduce the overall size of the array by dropping the last column on all matrices of `my_array` and store the result as `my_array_2`. Confirm the dimensions of the new array with a line of code.

Question 5

Predict the output of the following operation:

```
my_array[-6,,1] * diag(x=5)
```